Business Recommendations

Olist E-Commerce Performance Project | Improving customer retention (repeat rate: 3.12%)

Action 1: Encourage Second Purchases Through Targeted Incentives

Current Situation

The overall repeat rate is only **3.12%**. Repeat customers have a lower average order value (AOV) of approximately **122.80 BRL**, compared with **137.66 BRL** for non-repeat customers. However, repeat customers generate higher cumulative spending per customer: **382.93 BRL**, compared with **138.23 BRL** for non-repeat customers.

Root Cause / Hypothesis

The data suggests that repeat customers may generate more value through multiple purchases rather than higher spending per order. However, the current analysis does not establish why customers fail to return or whether discounts would increase retention.

Recommended Actions

- Test conditional incentives, such as a voucher for the second purchase or free shipping above a minimum order value, instead of offering blanket discounts.
- Personalize product recommendations based on the customer's first purchase and browsing or purchasing history, where available.
- For customers making high-value first purchases, prioritize complementary product recommendations and post-purchase engagement rather than assuming they need frequent discounts.
- Compare targeted campaigns with a control group to determine whether incentives increase repeat purchases enough to justify their cost.

Success Metrics

30-/60-/90-day second-purchase rate, incremental revenue per customer, average order value, and campaign cost per incremental repeat customer.

Action 2: Optimize Remarketing Timing and Category-Specific Recommendations

Current Situation

Repeat rates vary considerably by first-order category. Fashion Bags & Accessories records approximately **6.07%**, Furniture & Decor **4.88%**, Sports & Leisure **2.7%**, and Bed, Bath & Table **2.6%** in the earlier category analysis. Electronics-related categories have lower repeat rates, including Electronics at approximately **1.4%**, Auto at **1.6%**, and Computers & Accessories at **1.7%**. These figures should be validated against the final category definitions and model.

Root Cause / Hypothesis

Category differences may reflect different product replacement cycles, complementary-product opportunities, and customer needs. For example, customers buying durable products may have fewer reasons to purchase the same type of product again. Low repeat rates alone do not prove that the first purchase created a poor customer experience.

Recommended Actions

- Analyze time to second purchase using customer cohorts before selecting a remarketing window.
- Test follow-up campaigns at different intervals, such as 30, 45, 60, and 75 days after the first purchase, rather than assuming one window is optimal.

- For Fashion Bags & Accessories, Furniture & Decor, Sports & Leisure, and Bed, Bath & Table, test complementary products, coordinated bundles, and cross-selling.
- For Electronics, Auto, and Computers & Accessories, recommend compatible accessories, replacement components, or related products rather than relying only on repeat purchases of the same item.
- Use review and delivery data to personalize follow-up messages and avoid aggressive promotions for customers who have unresolved service issues.

Success Metrics

Second-purchase conversion rate by category, median time to second purchase, repeat revenue, cross-sell conversion rate, and campaign return on investment.

Action 3: Prioritize High-Volume Markets While Learning From High-Retention States

Current Situation

Customer distribution is concentrated in a few states. São Paulo (SP) has approximately **42K customers**, while Rio de Janeiro (RJ) has approximately **13K**. Meanwhile, AC records a repeat rate of approximately **5.19%**, and RO approximately **4.18%**, despite each having fewer than 1K customers in the displayed dashboard.

Root Cause / Hypothesis

High-volume states represent important markets because changes in their customer experience can affect many customers. Differences in repeat rates across states may reflect customer mix, category preferences, seller availability, or delivery performance. The current data does not establish that smaller states have better service capability or a more stable purchasing experience.

Recommended Actions

- Prioritize monitoring delivery reliability and seller performance in high-volume markets such as SP and RJ.
- Identify high-volume sellers in these states with elevated late rates or low review scores and investigate their operational issues.
- Analyze high-repeat states such as AC and RO to understand differences in category mix, customer behavior, seller distribution, and delivery performance.
- Compare states using both repeat rate and absolute repeat-customer count. Apply minimum sample-size thresholds before treating small-state results as reliable benchmarks.
- Test whether successful practices identified in high-retention states can be applied to larger markets.

Success Metrics

Repeat rate and repeat-customer count by state, late-delivery rate, average delivery days, review score, and customer retention campaign performance.

Action 4: Improve Seller Performance Through Delivery and Customer Experience Monitoring

Current Situation

Repeat customers have a higher late rate (**12.48%**) than non-repeat customers (**7.95%**), while average delivery duration is similar at **11.90 days** and **12.11 days**, respectively. Their average review scores are also similar: **4.12** for repeat customers and **4.08** for non-repeat customers. Seller-level data shows variation in late rates and review scores, including some high-revenue sellers with relatively low ratings.

Root Cause / Hypothesis

Average delivery duration may conceal differences in delivery reliability, including orders that arrive after the estimated date. Seller-level variation may also reflect differences in product mix, destination, order volume, or logistics arrangements. The higher late rate among repeat customers does not establish that late delivery increases or decreases the likelihood of repeat purchasing.

Recommended Actions

- Create a seller performance scorecard combining **Late Rate, Average Delivery Days, Average Review Score, Revenue, and Order Volume**.
- Prioritize sellers with both high order volumes and elevated late rates or low review scores, rather than ranking sellers by revenue alone.
- Investigate the causes of delays by seller, destination state, and product category before implementing operational changes.
- Track the percentage of orders delivered after the estimated date, not just average delivery duration.
- Analyze whether late delivery on a customer's first order is associated with a lower probability of making a second purchase, controlling for relevant customer and order characteristics.

Success Metrics

Late-delivery rate, average delivery duration, on-time delivery rate, average review score, seller-level performance improvement, and second-purchase rate following the first order.

Action 5: Strengthen Category and Seller Portfolio Management

Current Situation

Revenue is concentrated among leading product categories. Health & Beauty generates approximately **1.26M BRL**, Watches & Gifts **1.21M BRL**, and Bed, Bath & Table **1.04M BRL**. The dashboard also includes **3,095 sellers** and **74 product categories**. Seller performance varies, with some high-revenue sellers showing elevated late rates or lower review scores.

Root Cause / Hypothesis

Revenue concentration may expose the business to changes in demand, product availability, or seller performance in major categories. However, the top 10 categories generating a large share of revenue does not automatically indicate a problem; the business should also consider category profitability, growth, and customer demand.

Recommended Actions

- Measure each category's contribution to revenue, order volume, repeat rate, and growth to distinguish strong categories from those with retention or demand challenges.
- Investigate low-revenue categories using customer demand, order frequency, and seller performance before deciding whether to invest further or reduce assortment.
- Monitor high-revenue sellers using a balanced scorecard of revenue, order volume, late rate, and review score.
- Develop contingency plans for categories or sellers that contribute heavily to revenue, including monitoring product availability and identifying alternative suppliers where feasible.
- Explore cross-selling opportunities between leading categories and complementary lower-revenue categories when customer purchasing patterns support them.

Success Metrics

Revenue share of the top 10 categories, category-level revenue growth, seller revenue concentration, category repeat rate, late-delivery rate, and average review score.

Implementation and Evaluation

Prioritize these actions in the following order:

1. **Establish a reliable baseline:** Validate the repeat-rate definition and calculate second-purchase rates at 30, 60, and 90 days.
2. **Identify high-impact segments:** Segment customers by first-order category, order value, state, and first-order delivery experience.
3. **Run controlled experiments:** Test incentives, remarketing timing, and category-specific recommendations against control groups.
4. **Monitor seller performance:** Investigate high-volume sellers with elevated late rates or low review scores.
5. **Measure incremental impact:** Evaluate whether each intervention improves repeat purchasing and customer value after accounting for campaign costs.

Expected outcome: A more targeted customer retention strategy, improved delivery reliability, and better allocation of marketing and operational resources. Actual improvements should be confirmed through measurement rather than assumed in advance.